The Aesthetics of Generation

Do Large Language Models Have a Stylistic Fingerprint?

Project P13 – NLP, Università degli Studi di Milano

Abstract

Do LLMs have a recognisable "voice" that persists across prompts and genres? I generated a controlled corpus with four models on four genres, extracted 163 stylometric features with StyloMetrix, and analysed it with unsupervised and supervised methods. Unsupervised results showed that texts clustered by genre (adjusted Rand index 0.98) and not by model (-0.004): genre was the dominant source of stylistic variance, and the models separated only partially even within a single genre (ARI between 0.31 and 0.41). Nevertheless a Random Forest recognised the generating model with 89.9% accuracy in grouped cross-validation and 93.1% on a held-out test set built from unseen prompts (chance was 25%). The study then asked whether the fingerprint disappeared when a model tried to write differently. In the first test, I added a "mask" to the prompt, which is a short instruction that asks the model to change its style (for example to write like Hemingway, in an academic tone, in a casual tone, or like a teenager). The classifier wasn't retrained, and its accuracy fell to 70.3% but stayed far above chance, so the models still kept part of their signature when they tried to sound different. In the second test, I asked each model to imitate another model after reading two example texts of it. The imitation mostly failed: in three pairs out of four the classifier was fooled in at most 22% of the texts, and only GPT-OSS imitating Qwen worked in about half of the cases (53%). Even when they tried to copy someone else, the models seemed to carry their own stylistic signature with them.

1 Goal and research questions

The project investigated whether LLMs show a distinctive stylistic identity, following a recent study on the fingerprints of six LLMs on Japanese texts [1]. Previous work showed that machine-generated text can be detected [2] and that stylometric features help to tell AI-generated text from human text [3, 4]. I addressed four questions:

1. Do classic stylometric features separate the models?
2. Does unsupervised clustering group texts by model, or by something else (e.g. genre)?
3. Can a supervised classifier identify the generating model, also on prompts it has never seen?
4. How robust is this signature when the model is asked to change tone (masks) or to imitate another model?

2 Corpus

Models. Four models were queried through APIs (Groq and Mistral): Llama-3.3-70B (`llama`), Mistral Large (`mistral`), GPT-OSS-20B (`openai`) and Qwen3-32B (`qwen`). I used open-weight and freely accessible models because of cost; no closed frontier model such as GPT-4 or Claude was included.

Genres and roles. I used the four genres of the assignment: narrative, argumentation, dialogue and description. For each genre the models received a *role* through the system prompt, the same for all four models:

- **Narrative:** "You are a professional creative writer specializing in short stories."
- **Argumentation:** "You are an expert essayist specializing in formal and balanced argumentation."
- **Dialogue:** "You are a professional scriptwriter specializing in natural and realistic dialogue."
- **Description:** "You are a descriptive writer specializing in detailed environmental prose."

Only Qwen had one more sentence in the system prompt: "Respond ONLY in English. Do not use reasoning tags or other languages."

Prompts. Each request to a model was made of two parts: the *role* above (system prompt), which was the same for all the texts of a genre, and the *task* (user prompt), which was different every time. For each genre there were 50 tasks, written with the help of Claude, and each one was about a different subject. For example, two narrative tasks were *"Write a short story about an unexpected reunion..."* and *"Write a short story about discovering a forgotten..."*, and each one continued with the same kind of instructions on structure and tone (e.g. clear beginning, middle and end, neutral narrative tone). So the role fixed the genre, while the 50 tasks changed the content, and the texts didn't repeat the same story. Every task was given to all four models, so that content was balanced across models and couldn't be used to recognise them.

Settings. All models used the same settings: temperature 0.7, top-p 0.9, at most 1500 tokens. I set the limit of 1500 tokens to keep the generated texts in a similar range of length, so that length couldn't dominate the clusters or the classifier. The limit didn't make the lengths equal, but it avoided very long outputs.

Datasets. Table 1 summarises the datasets. The test set uses 10 *new* prompts per genre that were never seen during training.

Table 1: Datasets used in the project.

Dataset	Texts	Structure
Training	800	50 prompts × 4 genres × 4 models
Test (unseen prompts)	160	10 new prompts × 4 genres × 4 models
Masked (tone)	320	4 masks × 4 genres × 5 prompts × 4 models
Imitation (cross-model)	128	4 imitator→target pairs × 4 genres × 8 texts

3 Stylometric features

Features were extracted with StyloMetrix [5], which returns, for each text, a vector of normalised stylistic measurements. I used the 178 features that belonged to seven groups: part-of-speech proportions (`POS_`), lexical statistics such as type–token ratio and repetitions (`ST_`), syntactic patterns such as simple, coordinated and subordinate sentences (`SY_`), lexical and punctuation counts (`L_`), discourse/adverbial markers (`PS_`), verb tenses (`VT_`) and grammar such as the passive voice (`G_`). Missing values were set to 0, and I removed the 15 features that had zero variance, which left 163 features. The features were then standardised with the z-score (each feature had mean 0 and standard deviation 1, with mean and deviation computed on the training set and applied unchanged to the test texts), so that features with large values wouldn't dominate PCA, t-SNE, UMAP, K-Means and the Random Forest input.

4 Methods

Unsupervised analysis. The first approach was unsupervised, which means that the model labels weren't used. I reduced the features to two dimensions with PCA, t-SNE and UMAP to look at the texts as points on a plane, and I used all three methods to check that the picture doesn't depend on a single technique: they gave the same results. Then I ran K-Means with $k = 4$ (as many clusters as models and as genres) on the full feature space and on each of the three 2D spaces, to see how the texts were grouped. To measure how much the clusters agree with the true labels, I used the Adjusted Rand Index (ARI), which is 1 when the clusters match the labels perfectly and about 0 when the agreement is no better than chance. I repeated the same analysis inside each genre separately, to see if the models separated once the effect of the genre was removed.

Supervised analysis. A Random Forest (100 trees) predicted the generating model. Accuracy was measured with a 5-fold cross-validation whose folds were grouped by prompt (so that the same prompt never appeared in both training and validation) and, more importantly, on a separate test set with 160 texts from unseen prompts. I also reported precision, recall and F1 for each model, the confusion matrix, and the importance of the features (impurity based). A second forest used only the 40 most important features and is the one saved in the GitHub repository.

Robustness (tone masks). The 40-feature classifier was applied without retraining to texts generated with a style instruction added before the prompt: *Hemingway* (imitation of a human author), *academic*, *casual* and *Gen Z*. Masked texts used 5 of the training tasks per genre, and all genres shared the same role, "professional creative writer", instead of a role specific to the genre.

Imitation between models. For each target model and genre, two "prototypical" texts (those that the classifier attributes to the target with the highest probability) were given as examples to another model, which was asked to write a new text in their style. Each model acted as imitator for one target: llama→mistral, mistral→openai, qwen→llama and openai→qwen. The text was then classified: it counted as "fooled" if it was labelled as the target, "unmasked" if it was labelled as the imitator, and "other" otherwise.

5 Results

5.1 Genre dominates

In the unsupervised analysis I projected the 163 standardised features to two dimensions with three different methods (PCA, t-SNE and UMAP) and then ran K-Means on them. Figure 1 shows the same texts in the three projections, coloured by genre in the top row and by model in the bottom row. In every projection the texts form four well separated groups when they are coloured by genre, while the four models are mixed inside each group: the colours of the models overlap and no region belongs to a single model.

K-Means confirmed what the figure shows. When all genres were mixed, it recovered the four *genres* almost perfectly in every space (ARI 0.99 on all features, 0.98 on PCA, 0.97 on t-SNE, 0.98 on UMAP), while its agreement with the *models* was at chance level (ARI $= -0.004$ in all spaces). The three projections led to the same conclusion, so the result didn't depend on the choice of the method. To see whether the models separate once the effect of the genre is removed, I repeated the analysis inside each genre: I kept only the texts of one genre at a time, ran K-Means with $k = 4$ (one cluster for each model) and computed the ARI between the clusters and the models. The models started to separate even without labels, but only partly

(ARI 0.31 for argumentation, 0.33 for description, 0.39 for dialogue, 0.41 for narrative), far from the almost perfect agreement obtained for the genres.

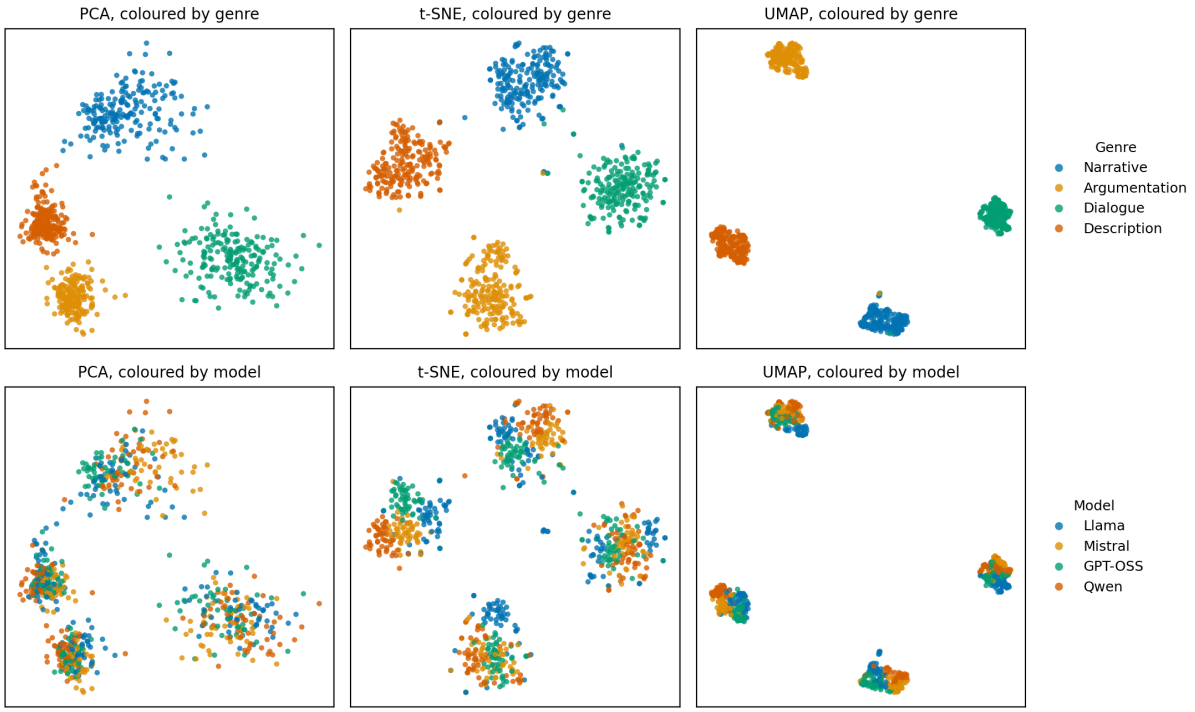


Figure 1: The 800 training texts projected with PCA, t-SNE and UMAP (columns). Top row: coloured by genre; bottom row: coloured by model. The genres form separate groups in all three projections, while the models overlap inside each group.

The two results are consistent. After standardisation, the genre shifts many features together by a large amount, so it carries much more variance than the model (on average the genre explains about 32% of the variance of a feature and the model about 4%), which is exactly what PCA and K-Means pick up. The model effect is smaller and concentrated in a few features. A supervised method such as the Random Forest chooses the features that separate the labels regardless of their variance. The model signature therefore lives in a lower-variance direction, hidden by the genre.

5.2 Model identification

Table 2: Random Forest accuracy (chance = 25%).

Setting	Accuracy
5-fold CV grouped by prompt, 163 features	89.9% (folds from 86.9% to 92.5%)
Test set, unseen prompts, 163 features (149/160)	**93.1%** (95% CI 88.1–96.1)
Test set, top-40 features only	90.0%

The supervised approach used the labels. I trained a Random Forest with all the 163 features and measured its accuracy in two ways; Table 2 summarises the results of all the settings. In the cross-validation the folds were grouped by prompt, so the classifier couldn't exploit the content of a prompt it had already seen, and the accuracy was 89.9%. On the test set, which contained completely new prompts, the accuracy was 93.1% (149 texts out of 160), in line with the cross-validation, and its 95% confidence interval went from 88.1% to 96.1%, far above the chance level of 25%. By genre, the accuracy on the test set went from 87.5% (description) to

97.5% (dialogue). Figure 2 shows the confusion matrix, which tells which models are confused with each other, and the precision, recall and F1 of each model: all these values are reported in the figure.

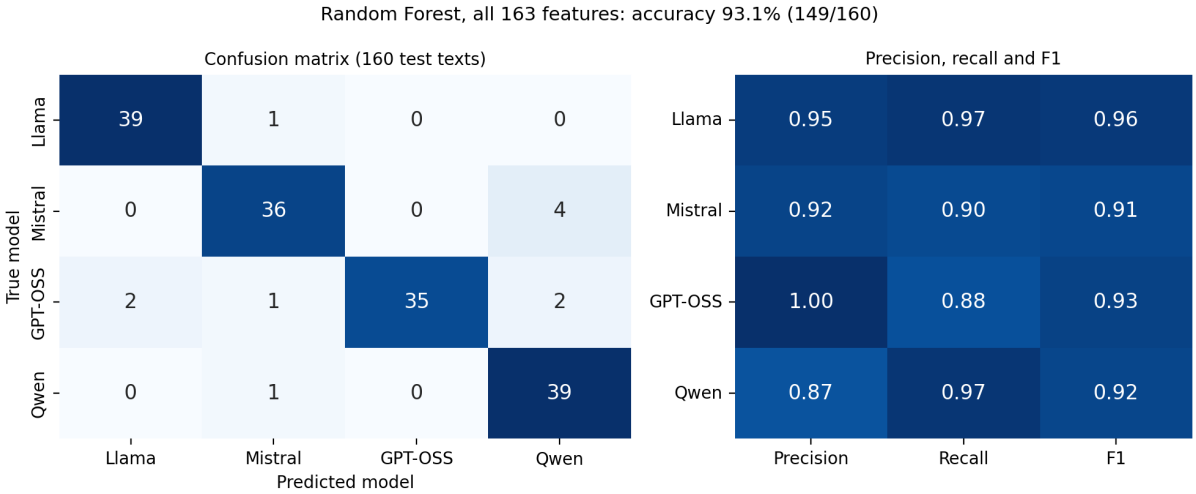


Figure 2: Random Forest with all the 163 features on the test set (160 texts, unseen prompts). Left: confusion matrix (rows are the true models, columns the predicted ones). Right: precision, recall and F1 for each model.

From the forest trained on the 163 features I computed the feature importance (the mean decrease of impurity, which measures how much each feature helps the trees to separate the models), computed on the training set only. The 40 most important features came out as the most discriminant ones, and I decided to keep only those, which is about a quarter of the original features. I then trained a new Random Forest on these 40 features, with the same settings. The accuracy on the test set was 90.0% (144 texts out of 160), and Figure 3 reports the confusion matrix and the precision, recall and F1 of each model. Using only 40 features cost just 3.1 points (from 93.1% to 90.0%, which is 5 texts out of 160), and the simpler model was preferable: it was easier to interpret, it was the one used in the stress tests of the next sections, and it showed that the signature is concentrated in few features. The most informative ones were the use of dashes (`L_PUNCT_DASH`), the type–token ratio of lemmas, the number of content words, the repetition of words, and the overall punctuation density, followed by function words, colons, conjunctions, simple sentences and commas. These are the concrete "tics" that make up the fingerprint. In conclusion, the classifier was able to tell with good accuracy which model is behind a generated text, even with a reduced set of features.

5.3 Robustness to tone

Table 3: Accuracy of the (non-retrained) top-40 classifier on masked texts, 80 texts per mask.

Mask	Accuracy
Casual	81.2%
Academic	71.2%
Gen Z	70.0%
Hemingway	58.8%
Overall	70.3%

Table 3 reports the accuracy for each mask. The signature degraded under a stylistic instruction (from 90.0% to 70.3%) but stayed far above chance, which suggests a stylistic core

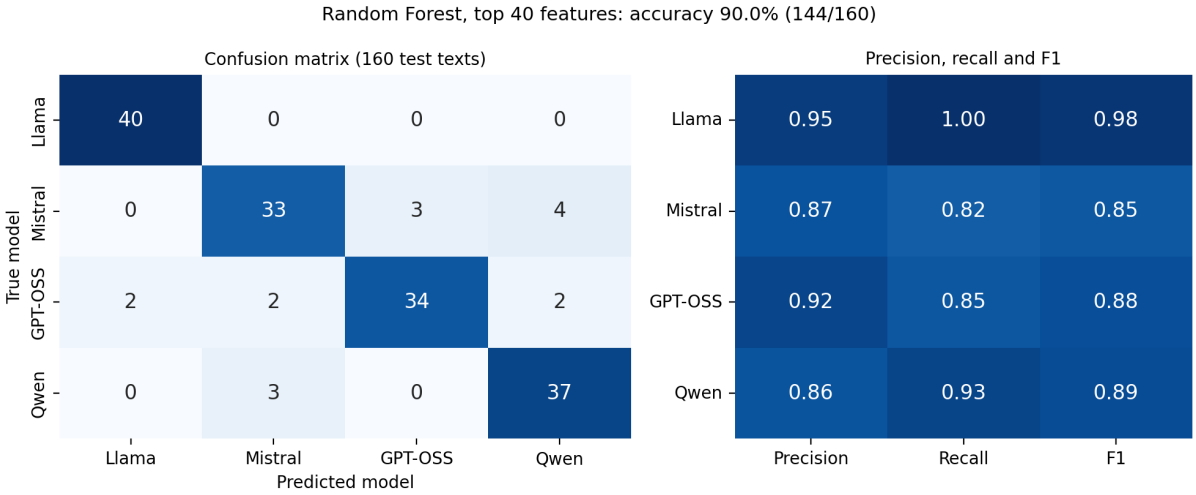


Figure 3: Random Forest with the 40 most important features on the test set. Left: confusion matrix. Right: precision, recall and F1 for each model.

that survives a change of tone. Imitating a human author (Hemingway) was the strongest disguise, and the casual tone was the weakest. The masked prompts came from the training set and used a generic system prompt, so the drop mixed the effect of the mask with the change of system prompt. There was no control condition without the mask (see limitations).

5.4 Imitation between models

Table 4: Imitation of a target model by another model (32 texts per pair, in %).

Imitator → target	Fooled (labelled as target)	Unmasked (labelled as imitator)	Other
llama → mistral	21.9	65.6	12.5
mistral → openai	21.9	68.8	9.4
qwen → llama	3.1	78.1	18.8
openai → qwen	53.1	37.5	9.4

Table 4 shows, for each pair, how the classifier labelled the imitated texts. In three pairs out of four the imitator was recognised as itself in 66–78% of the cases and fooled the classifier in at most 22%, and Qwen almost never managed to pass as Llama (3.1%). Only GPT-OSS imitating Qwen succeeded in more than half of the cases. Looking at the features, the texts written by GPT-OSS when imitating Qwen lay between the two models: they moved towards Qwen but didn't reach it. So the classifier could be fooled even when the style didn't really match the target. Two examples weren't enough to "wear" another model's statistical signature.

6 Limitations

- **Models.** Only four models were used, all accessed through public APIs; no closed frontier model (GPT-4, Claude) was included, and the original models can no longer be queried to repeat the generation.
- **Imitation prompts.** In the imitation data the 32 texts of each pair came from only 2 distinct prompts per genre, repeated several times, so the percentages are only indicative.
- **Single feature extractor.** Results depended on StyloMetrix (English) and on impurity-based feature importance, which could be replaced by permutation importance or SHAP, as done by Zaitsu et al. [1].

7 Conclusion

This study showed that the four LLMs analysed leave a measurable stylistic fingerprint in the texts they generate. The genre dominated the variance, so the unsupervised methods grouped the texts by genre and not by model, but a Random Forest recovered the fingerprint with 89.9% accuracy in cross-validation and 93.1% on unseen prompts (90.0% with only 40 features). The fingerprint was also robust: with tone masks the accuracy fell to 70.3% but stayed far above chance, and imitating another model mostly failed. These findings suggest that the texts can be distinguished by the combination of many small stylometric features, even though all four models share the same Transformer architecture.

References

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